MAHATMA GANDHI UNIVERSITY, KOTTAYAM

FIRST SEMESTER MGU- UGP (HONOURS) REGULAR EXAMINATION NOVEMBER 2024

Multi- Disciplinary Course – MG1MDCBTG100- ECOLOGY AND ENVIRONMENTAL SCIENCE

(2024 ADMISSION ONWARDS)

Scheme for Valuation

Duration: 1 Hours **Maximum Marks: 35**

Part A

Answer the Questions in One Word or Sentence

Answer all questions.

Each question carries 1 mark

1 .Which trophic level occupies the base of the ecological pyramids [K] [1]

Answer: Producers

2 .Which group in an ecosystem includes bacteria and fungi [U] [1]

Answer: Decomposers

3. Name any one national park in India. [A] [2]

Answer: Any one

4 .A set of ecosystem is called ..... [K] [1]

Answer: Biome

5 .What is the apex of an ecological pyramid? [K] [1]

Answer: Tertiary consumers

6 .Give an example for abiotic component in an ecosystem. [U] [1]

Answer: Any one

7 .Which ministry oversees the implementation of the NRCP? [U] [3]

Answer: Ministry of Environment and Forest

8 .What climate challenge involves glacial melt and affects the region? [A] [3]

Answer: Global Warming

9 .What is the primary focus of CAMPA's fund allocation? [A] [3]

Answer: Protection of Forest

10. What tree species is often planted in afforestation projects in India? [K] [3]

Answer: Any commonly planted tree

(10 × 1 = 10)

Part B

Short Answer Questions-Answer any 3 questions.

Each question carries 3 marks

11. Define a food chain and explain its importance in ecosystem. [U] [1]

Answer: Food chain- definition- 1 mark, importance- 2 points- 2 marks

12. Illustrate how afforestation and reforestation efforts under the Green India Mission can impact local ecosystems and communities. [A] [3]

Answer: Green India Mission – impacts-Any 3 points-3 marks

13. Differentiate between Beta diversity and Gamma diversity. [A] [2]

Answer: Biodiversity- definition- 3 marks (types not specified in the syllabs)

14. Explain the significance of biodiversity in the Sundarbans mangrove ecosystem. [U] [3]

Answer: Sundarbans mangrove ecosystem – 2 marks, significance- 1 mark

15. What are the major types of ecosystem? [K] [1]

Answer: types of ecosystem with examples – 3 marks **(3 × 3 = 9)**

Part C

Short Essay Questions- Answer any 1 question.

Each question carries 6 marks.

16. Explain the ecological importance of the Nilgiri Tahr and its role in the ecosystem. [A] [3]

Answer: Nilgiri Tahr- introduction – 2 marks, Ecological importance (minimum 4 points)- 4 marks

17. Explain the activities involved in germplasm conservation. [A] [2]

Answer: Germplasm conservation -Definition- 2 marks, Insitu and Exsitu conservation - any 4 Strategies- 4 marks **(1 × 6 = 6)**

Part D

Essay Questions-Answer any 1 question (10 marks)

18**.** Compare and contrast different types of food chains in detail. [U] [1]

Answer: food chain- Definition- 2 marks, components of a typical food chain- 6 marks, types -2 marks

19 .Evaluate the effectiveness of the "Green India Mission" in enhancing forest cover and biodiversity. [A] [3]

Answer: "Green India Mission"- introduction- 1 mark, Objectives- 5 marks, Effectiveness/ implementation- 4 marks.

(1 × 10 = 10)

END OF THE QUESTION PAPER